



ACID, BASE & SALT — FULL MIND MAP

Chemistry · Chapter 6.D · Topic-wise · Fully Explained · Hinglish · SSC/PCS/UPSC Ready

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1. Acids, Bases, Antacids & Aqua Regia

- Antacids basic nature ke compounds hote hain — jaise $\text{Mg}(\text{OH})_2$, $\text{Al}(\text{OH})_3$, CaCO_3 etc. Ye stomach mein excess HCl ko neutralise karke acidity/heartburn ko reduce karte hain. [Q1]
- Aqua regia = concentrated HCl : concentrated HNO_3 ka 3:1 mixture. Ye strongly corrosive/oxidising mixture hai aur Gold, Platinum jaise noble metals ko dissolve kar sakta hai. [Q2]

2. pH Scale, Litmus & Hydronium-Ion Concept

- pH aqueous solution ki acidity/basicity ko show karta hai: pH < 7 acidic, pH = 7 neutral aur pH > 7 basic/alkaline. Pure water aur ideal neutral solution ka pH 7 hota hai. [Q3, Q5, Q6, Q10]
- pH logarithmic scale hai: pH 3 se 6 hone par H^+ concentration $10^3 = 1000$ times decrease hoti hai. Isliye pH mein chhota numerical change bhi ion concentration mein large change show karta hai. [Q4]
- Base red litmus ko blue karta hai; acid blue litmus ko red karta hai. Distilled water neutral hone ki wajah se litmus ka colour change nahi karta. Red litmus blue ho to pH definitely 7 se greater hona chahiye. [Q7, Q8]
- Dry HCl gas dry litmus ka colour nahi badalta kyunki water absent hone par H_3O^+ /hydrated proton form nahi hota. Acidic behaviour aqueous medium mein express hota hai. [Q9]



EXAM TRAP — Q7 mein correct concept option-list se bahar hai

- ▶ Red litmus → blue ka matlab solution basic hai, yani pH > 7. Given options 7, 1, 5, 6.5 the, isliye source ne "None of the above" key kiya.
- ▶ Concept yaad rakho: indicator-result ko directly pH range se map karo; option-number ko fact mat samjho.

3. Industrial Acids & Acid Rain

- Sulphuric acid (H_2SO_4) ko "basic chemical of industry" kaha jata hai, kyunki fertilizer, petroleum refining, metal treatment aur large-scale chemical manufacture mein iska extensive use hota hai. [Q11]
- Acid rain mainly atmospheric SO_2 aur nitrogen oxides ke oxidation se banne wale H_2SO_4 aur HNO_3 se associated hoti hai. Given-option questions mein Sulphuric acid ko major component aur Nitric acid ko bhi principal acid maana gaya hai. [Q12, Q13, Q14]



EXAM TRAP — Acid-rain composition fixed percentage nahi hoti

- ▶ Source Q12 H_2SO_4 ko ~70–75% batata hai; actual acid-rain composition place, season aur emissions ke according vary karti hai.
- ▶ Exam-safe core: major mineral acids = Sulphuric acid + Nitric acid.

4. Salt Hydrolysis, Lewis Concept & Safe Dilution

- CuSO_4 jaise salt ka aqueous solution acidic ho sakta hai because cation hydration/hydrolysis H^+ generate karne wali equilibrium create karti hai. Exam answer: hydrolysis. [Q15]
- Lewis acid electron-pair acceptor aur Lewis base electron-pair donor hota hai. BF_3 , AlCl_3 , FeCl_3 electron-pair acceptors hain; NH_3 ke paas lone pair hota hai, isliye NH_3 Lewis base hai. [Q16]
- Concentrated H_2SO_4 ko dilute karte waqt acid ko slowly water mein add karte hain — "Always Add Acid". Dilution highly exothermic hai; water ko acid mein daalne par local boiling/splashing ka danger hota hai. [Q17]



EXAM TRAP — Q15 ki literal molecular equation ko mechanism mat samjho

- ▶ Source $\text{CuSO}_4 + 2\text{H}_2\text{O} \rightarrow \text{H}_2\text{SO}_4 + \text{Cu}(\text{OH})_2$ likhta hai; aqueous salt hydrolysis ko is simple one-way molecular equation se represent karna oversimplified hai.
- ▶ Exam point: CuSO_4 solution acidic due to hydrolysis/hydrated Cu^{2+} acidity.

5. Acid–Base Reactions, Amphoteric Oxides & CO₂ Chemistry

- Amphoteric oxides acid aur base dono ke saath react kar sakte hain. Classic examples: Al₂O₃ aur ZnO. CuO mainly basic oxide hai. [Q18]
- Acid + metal carbonate → salt + H₂O + CO₂. Isi CO₂ ko limewater [Ca(OH)₂] mein pass karne par insoluble CaCO₃ banta hai aur limewater milky ho jata hai. [Q19, Q20]
- CO₂ water mein dissolve hokar carbonic acid (H₂CO₃) form karta hai, isliye soda water mildly acidic hota hai. HCl oxygen-free mineral acid hai; HNO₃ aur H₂SO₄ oxygen-containing acids hain. [Q21, Q22]
- Ammonia water mein dissolve hokar basic solution deta hai aur red litmus blue karta hai. HCl gas ke saath NH₃ dense white fumes of NH₄Cl banata hai — unknown-gas identification ka classic test. [Q23]

⚠ EXAM TRAP — “NH₄OH” ko convenient school-level notation samjho

- Aqueous ammonia ka better equilibrium representation: $\text{NH}_3 + \text{H}_2\text{O} \rightleftharpoons \text{NH}_4^+ + \text{OH}^-$.
- Source “ammonium hydroxide” wording use karta hai; exam mein expected basic behaviour aur NH₄Cl white fumes yaad rakho.

6. Common Salts — Baking Soda, POP, Blue Vitriol & Matchings

- Baking soda = Sodium bicarbonate / Sodium hydrogen carbonate = NaHCO₃. Baking mein leavening agent ke roop mein CO₂ release karta hai; Q24 aur Q32–Q35 isi formula/name ko repeat karte hain. [Q24, Q32, Q33, Q34, Q35]
- Important matching: Bleaching powder → Calcium oxychloride/hypochlorite; Baking soda → NaHCO₃; Plaster of Paris → CaSO₄·½H₂O. POP ko Calcium bicarbonate kehna wrong hai. [Q25, Q26]
- Neela thotha / Blue vitriol = hydrated Copper sulphate, commonly CuSO₄·5H₂O. Copper sulphate fungicide ke roop mein water tanks/crops mein use ho sakta hai. [Q27, Q28, Q29]
- High-yield salt map: Washing soda → Sodium carbonate; Caustic soda → NaOH; Neela thotha → Copper sulphate; Hypo → Sodium thiosulphate; Epsom salt → Magnesium sulphate; Baking soda → NaHCO₃. [Q30, Q31]

⚠ EXAM TRAP — Q25 mein “washing powder” wording simplified hai

- Source matching “washing powder → sodium carbonate” key karta hai. Commercial washing powders mixtures hote hain; sodium carbonate unka builder/component ho sakta hai.
- Washing soda ko specifically Na₂CO₃·10H₂O se distinguish karo.

7. Washing Soda, Common Salt, Iodised Salt & Bleaching Powder

- Washing soda ka chemical base Sodium carbonate hai; hydrated washing soda = Na₂CO₃·10H₂O, jabki anhydrous Na₂CO₃ ko soda ash kaha jata hai. Q36–Q37 name/formula recall karte hain. [Q36, Q37]
- Baking soda fire-extinguisher mixtures mein CO₂ source ho sakta hai. Bleaching powder historical industrial processes such as Hasenclever/Backmann plants se prepare kiya gaya; source Q38 mein statements 2–3 correct hain, statement 1 wrong. [Q38]
- Ocean water ka predominant dissolved salt NaCl hai. Table salt strong acid HCl + strong base NaOH ka neutral salt hai. [Q41, Q42]
- Iodised salt, sea salt, rock salt aur processed common salt sab dining/table-use forms ho sakte hain; source Q43 mein koi unique “should not be used” option establish nahi hota, isliye ise disputed mark kiya gaya hai. [Q43]
- Iodised salt iodine-fortified common salt hota hai; iodising agents iodide/iodate salts ho sakte hain. Source option-set Q44 Potassium iodide ko correct key karta hai. [Q44]
- Gypsum cement ki setting ko slow/control karne ke liye add kiya jata hai. Ba aur Sr highly reactive alkaline-earth metals hain, isliye nature mein free state mein nahi milte. [Q45]
- Bleaching powder oxidising/bleaching agent hai; dilute acid se chlorine release kar sakta hai aur water treatment mein use hota hai. Main exam formula often CaOCl₂ / calcium oxychloride likha jata hai. [Q46, Q47]

⚠ EXAM TRAP — Q43 disputed hai; Q44 source-specific iodisation chemistry hai

- Q43 ke listed iodised, sea, rock aur processed common salt sab edible/table-use forms ho sakte hain; source ne question ko disputed (*) maana hai.
- Iodised salt ke liye practical fortificants mein iodide aur iodate dono milte hain; exact exam option/source wording follow karo.

8. ENO, Photography Chemicals — Hypo & Silver Bromide

- ENO-type effervescent salt mein commonly sodium bicarbonate + citric acid react karke CO₂ bubbles produce karte hain. Isi effervescence se fizzy action hota hai. [Q48]
- Photography mein “Hypo” = Sodium thiosulphate solution. Ye fixer/antichlor ke roop mein unreacted silver halide ko soluble complex bana kar remove karta hai. [Q49, Q50, Q51, Q52, Q53]
- Photographic films/plates ki light-sensitive emulsion mein Silver bromide (AgBr) classic photosensitive salt hai; light exposure latent image banane ki process initiate karta hai. [Q54, Q55]

⚠ EXAM TRAP — Developer aur fixer ko mix mat karo

- ▶ Source Q49 wording "develop photographic plates" use karta hai, lekin Sodium thiosulphate ka standard role FIXER hai, developer nahi.
- ▶ Q54 "useful element" bolta hai but Silver bromide (AgBr) element nahi, compound hai. Exam intent photosensitive photographic salt ko identify karna hai.

9. Quartz, Pottery, Coloured Glass & Ionic Compounds

- Source Q39 matching mein Artificial ripening → Ethylene, Baking soda → NaHCO_3 , Soft drink → CO_2 aur Quartz → Sodium silicate pair diya gaya hai; lekin quartz ka actual chemical composition SiO_2 hai, jise Q40 directly test karta hai. [Q39, Q40]
- Pottery glazing mein ZnO use kiya ja sakta hai; glass ko deep blue colour dene ke liye Cobalt oxide classic colouring agent hai. [Q56, Q57]
- Ionic compounds generally organic solvents like alcohol mein poorly soluble hote hain. Solid ionic crystal mein ions fixed lattice positions par hote hain, isliye solid state mein electricity conduct nahi karte; molten ya aqueous state mein conduct karte hain. [Q58]

⚠ EXAM TRAP — Q39 Quartz matching ko direct chemistry fact mat banao

- ▶ Quartz = SiO_2 (silicon dioxide). Source Q39 mein Sodium silicate pair elimination/context se aata hai; Q40 correct direct composition SiO_2 confirm karta hai.
- ▶ Q58 ka "insoluble in alcohol" general exam-rule hai; individual ionic compounds ki exact solubility solvent aur salt par depend kar sakti hai.

✓ Source-question mapping preserved: Q1–Q58